

OM SALUNKE

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SUMMARY

Results-driven AI and Software Engineer holding a Master's degree in Artificial Intelligence from the University at Buffalo. Proven expertise in architecting scalable AI data pipelines, RAG frameworks, and multi-agent systems alongside cloud deployments. Adept at bridging complex ML models with production-ready applications.

EDUCATION

Masters in Artificial Intelligence | University at Buffalo, NY

January 2026

SKILLS

Programming & Web: Java, Python SQL, C++, ReactJS, Redux, RESTful APIs, Git/Version Control, Web3.js

AI & LLM Engineering: RAG Architectures, MCP, Prompt Engineering, LangGraph, AutoGen, Vector Databases

Cloud & Deployment: GCP, AWS, Azure, Vercel, Terraform, Docker, GitHub Actions, CI/CD Pipelines

MLOps & Testing: Databricks, PySpark, Unity Catalog, MLflow, JUnit, Mockito, Selenium, Agile Methodologies

WORK EXPERIENCE

Research Assistant – University at Buffalo

Present

- **Led end-to-end AI implementation** by developing asynchronous **Python** and **FastAPI microservices** to power **Agentic AI** workflows, routing 100K+ medical records through **AWS Bedrock** foundation models with a 30% reduction in inference latency.
- Designed highly scalable **RAG pipelines and ML data layers** using Databricks Delta Lake, implementing strict Unity Catalog access controls and **MCP** to safely expose 5TB+ of enterprise context to language models with zero security breaches.

Software Engineer – Tech Mahindra

June 2023

- Architected enterprise-grade **blockchain networks** utilizing **Hyperledger** and **AI-augmented** smart contracts to model digital assets, leveraging **machine learning algorithms** to securely route and execute cross-organizational transactions across distributed nodes.
- Built scalable frontend **DApp components** integrated with Web3.js for secure wallet handling, deploying **predictive ML models and AI-driven anomaly detection** to continuously monitor system stabilization and autonomously resolve technical incidents for global high availability.

PROJECTS

Ripple - AI Policy Simulator ([GitHub](#))

- Architected a scalable data pipeline and simulation engine using **Python**, **AWS**, and **Databricks (PySpark, Delta Lake)**, processing 100K+ Monte Carlo simulations 40% faster for predictive economic modeling.
- Engineered an interactive visualization dashboard utilizing **ReactJS** and the **Google Gemini API**, empowering users to dynamically adjust 20+ economic parameters with under 1s latency.

ShopSphere: Production-Grade eCommerce Platform Development ([GitHub](#))

- Demonstrated strong proficiency in Java, utilizing Spring Boot, Spring Security, and Hibernate ORM to engineer a scalable eCommerce backend API, **architecting robust JUnit and Mockito test suites** to validate complex business logic and ensure high-performance data retrieval.
- Engineered a responsive full-stack platform by integrating a React-Redux frontend with Java-driven REST endpoints, establishing strict JWT-based access controls and **executing end-to-end automated API testing**.